The goal of this experiment was to implement **Named Entity Recognition (NER)** using Python and spaCy.
NER is a core task in Natural Language Processing (NLP) that identifies and classifies key elements in text into predefined categories such as **names of people, organizations, locations, dates, etc.**

2. Tools and Libraries Used

- **Python 3 (Google Colab)** – for implementation
- **spaCy** – for performing NER using its pre-trained English language model
- **Pandas** – for displaying extracted entities in a tabular format
- **displacy** – for visualizing entities in colored highlights

3. Steps Performed

Step 1: Installing and Loading spaCy

!pip install spacy

import spacy

nlp = spacy.load("en_core_web_sm")

The pre-trained **en_core_web_sm** model (English small model) was loaded. It includes:

- Tokenizer
- Part-of-speech tagger
- Named Entity Recognizer

Step 2: Input Text for Analysis

A few example texts were tested:

text = "Elon Musk founded SpaceX in 2002 and Tesla Motors in 2003."

and later:

user_text = "Barack Obama was born in Hawaii and served as the President of the United States."

Step 3: Performing NER

The text was processed using spaCy's NLP pipeline:

```
doc = nlp(user_text)

for ent in doc.ents:

   print(ent.text, ent.label_)
```

This step extracted entities such as:

Barack Obama → PERSON

Hawaii → GPE

United States → GPE

Step 4: Visualizing Entities

Using **displacy**, entities were displayed with colored highlights for better understanding:

```
from spacy import displacy

displacy.render(doc, style="ent", jupyter=True)
```

Step 5: Tabular Output

To make the results more readable:

```
import pandas as pd

entities = [(ent.text, ent.label_) for ent in doc.ents]

pd.DataFrame(entities, columns=["Entity", "Label"])
```

Entity	**Label**
Barack Obama	PERSON
Hawaii	GPE
the United States	GPE

4. Model Choice

The **spaCy en_core_web_sm** model was chosen because:

- It is **lightweight and fast**, suitable for academic and prototype projects.
- It comes **pre-trained** on a large English corpus (OntoNotes 5).
- It provides a complete **NLP pipeline** (tokenization, tagging, parsing, and NER).
- It requires **no GPU or heavy setup** — ideal for running smoothly in **Google Colab**.

Compared to other models like transformer-based **BERT**, spaCy is:

- Easier to use for beginners,
- Requires less computational power,
- Sufficient for standard NER tasks and demos.

5. Observations

- spaCy efficiently detected **persons**, **organizations**, and **locations**.
- For generic sentences (like "Hi, how are you?"), no entities were detected — which is correct behavior.
- Visualization clearly highlighted recognized entities, making the output easy to interpret.

6. Conclusion

This project successfully demonstrated how **Named Entity Recognition (NER)** can be performed using the **spaCy** library in Python.
The experiment showed that spaCy's pre-trained model is capable of accurately detecting and classifying entities in natural text with minimal setup and excellent performance.

In summary:

- Implemented NER using spaCy
- Visualized entities interactively in Colab
- Displayed clean, tabular results
- Explained model choice and outcomes

7. Future Scope

- Integrate NER with real-world datasets (e.g., news articles or resumes).
- Compare spaCy's results with **transformer-based models (BERT or RoBERTa)** for improved accuracy.
- Deploy the NER model as an API or a small web app for automatic information extraction.